

20 Essential **LLM** **Guardrails**

For Safe and Responsible AI Use



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Anti-Prompt Injection Defense

Security and Privacy

- ✓ Blocks Jailbreaks: detect and stop attempts to trick the model into breaking rules.
- ✓ Sanitizes Inputs: Neutralizes hidden commands or malicious instructions in user prompts.
- ✓ Protects System Prompts: Ensures the core identity and instructions of the AI remain intact.
- ✓ Prevents Manipulation: Stops users from overriding safety protocols via clever phrasing.



Hallucination Prevention & Fact-Checking

Response and Relevance

- ✓ **Verifies Claims:** Real-time cross-referencing of generated facts with trusted knowledge bases.
- ✓ **Reduces Confabulations:** Filters out plausible-sounding but incorrect information.
- ✓ **Cites Sources:** Ensures every claim is backed by a verifiable reference or document.
- ✓ **Grounds Responses:** Forces the model to use provided context rather than making things up.



Sensitive Data Screening (PII)

Security and Privacy

- ✓ Redacts Personal Info: Automatically masks names, phones, emails, and addresses.
- ✓ Prevents Leakage: Stops the model from inadvertently revealing training data secrets.
- ✓ Complies with Privacy Laws: Helps adhere to GDPR, CCPA, and data protection standards.
- ✓ Filters Financial Data: Blocks credit card numbers and banking details from outputs.



Language Toxicity Detection

Security and Privacy

- ✓ Filters Hate Speech: Blocks slurs, discriminatory remarks, and harassment.
- ✓ Monitors Tone: Ensures interactions remain polite and professional.
- ✓ Detects Aggression: Identifies and neutralizes hostile or threatening language.
- ✓ Safe for Work: Guarantees content is appropriate for professional environments.



Contextual Relevance Check

Response and Relevance

- ✓ **Stays on Topic:** Ensures the answer directly addresses the user's specific question.
- ✓ **Filters Noise:** Removes tangential or unrelated information from the response.
- ✓ **Validates Coherence:** Checks if the response logically follows the conversation history.
- ✓ **Prevents Drift:** Stops the model from wandering into unrelated subjects.



SQL Injection Prevention

Logic and Functionality Validation

- ✓ Sanitizes Queries: Detects and blocks malicious SQL code in natural language inputs.
- ✓ Validates Syntax: Ensures generated SQL is syntactically correct and safe to run.
- ✓ Restricts Scope: Prevents unauthorized database access or modification attempts.
- ✓ Protects Data Integrity: Shields your database from destructive commands.



Competitor Shield & Brand Safety

Content Validation and Integrity

- ✓ Blocks Competitor Mentions: Prevents the AI from promoting or discussing rival products.
- ✓ Enforces Brand Tone: Ensures responses align with your company's voice and values.
- ✓ Filters Negative Sentiment: protecting your brand reputation in generated text.
- ✓ Controls Recommendations: strictly suggests your own services and solutions.



Link Validity Checker

Response and Relevance

- ✓ Tests URLs: Pings generated links to ensure they are live and not 404s.
- ✓ Prevents Dead Ends: Removes or replaces broken links before showing them to users.
- ✓ Verifies Safety: Checks linked sites against malicious domain lists.
- ✓ Ensures Accessibility: Guarantees resources shared are actually reachable.



Intent Alignment Verification

Response and Relevance

- ✓ Analyzes Goal: Confirms the response actually solves the user's underlying problem.
- ✓ Checks Instruction Following: Verifies if constraints (e.g., 'answer in 3 bullets') were met.
- ✓ Prevents Misunderstandings: Clarifies ambiguous queries before answering.
- ✓ Ensures Satisfaction: Aligns output format and depth with user expectations.



Structured Data/JSON Validator

Logic and Functionality Validation

- ✓ Enforces Schema: Checks if generated JSON matches the required structure/keys.
- ✓ Fixes Syntax: Auto-corrects missing braces, quotes, or commas.
- ✓ Type Checking: Ensures values (strings, ints) matched expected data types.
- ✓ Parsability Guarantee: Ensures downstream systems can read the AI's output without crashing.



Multilingual Accuracy Audit

Language Quality

- ✓ Preserves Nuance: Ensures translations keep the original meaning and tone.
- ✓ Checks Grammar: Validates syntax and vocabulary in the target language.
- ✓ Cultural Context: Adapts idioms and references for the specific audience.
- ✓ Prevents Mistranslation: Flags potential errors in critical communications.



Coherence & Logic Filter

Content Validation and Integrity

- ✓ Detects Gibberish: Filters out repetitive or nonsensical text loops.
- ✓ Checks Argument Flow: Ensures premises logically lead to conclusions.
- ✓ Removes Hallucinated Words: Spots non-existent words or garbled syntax.
- ✓ Quality Assurance: Guarantees a polished, human-readable output.



Complexity & Readability Scoring

Language Quality

- ✓ Adjusts Level: Measures Flesch-Kincaid score to match audience expertise.
- ✓ Simplifies Jargon: Explains technical terms for general audiences.
- ✓ Enhances Clarity: Refines sentence structure for maximum readability.
- ✓ Tone Matching: Switches between formal/technical and casual/simple styles.



Dynamic Pricing Verification

Content Validation and Integrity

- ✓ Real-Time Lookup: Validates quoted prices against current database values.
- ✓ Prevents Hallucinated Discounts: Stops the AI from inventing fake deals.
- ✓ Currency Check: Ensures correct currency symbols and formatting.
- ✓ Inventory Sync: Only quotes prices for items actually in stock.

API Schema Compliance

Logic and Functionality Validation

- ✓ **Validates Rest Calls:** Checks if generated API requests match the OpenAPI spec.
- ✓ **Parameter Check:** Ensures required fields are present and valid.
- ✓ **Method Enforcement:** Verifies correct HTTP methods (GET, POST, etc.) are used.
- ✓ **Endpoint Accuracy:** Prevents calls to non-existent or deprecated API endpoints.



Citation & Source Tracing

Content Validation and Integrity

- ✓ Attribution Check: Ensures quotes are correctly attributed to their authors.
- ✓ Link Verification: Connects claims directly to the source document.
- ✓ Prevents Plagiarism: Rewrites content that is too similar to training data.
- ✓ Audit Trail: Provides a clear path of where information came from.



Redundancy Removal

Language Quality

- ✓ De-duplicates Ideas: Merges repetitive points into concise statements.
- ✓ Trims Fluff: Removes filler words that add no value.
- ✓ Conciseness Optimization: Shortens content without losing meaning.
- ✓ Streamlines Output: Creates punchy, direct responses.



Output Quality Assessment

Language Quality

- ✓ Grades Relevance: Scores how well the answer fits the prompt.
- ✓ Evaluates Completeness: distinct score for missing information.
- ✓ Feedback Loop: Uses low scores to trigger retries or human review.
- ✓ Quality Standard: Enforces a minimum quality bar before showing the user.

Logical Consistency Checker

Logic and Functionality Validation

- ✓ Spot Contradictions: identifying self-conflicting statements in a single response.
- ✓ Temporal Consistency: Ensures timelines and dates make chronological sense.
- ✓ Entity Consistency: Keeps names and roles consistent throughout the text.
- ✓ Argument Integrity: logic flow validation.



Inappropriate Content Filter

Security and Privacy

- ✓ NSFW Filter: Strictly blocks adult or explicit content generation.
- ✓ Violence Detection: Stops generation of gory or graphically violent text.
- ✓ Self-Harm Prevention: Intervenes if user prompts suggest self-harm.
- ✓ Safety First: The baseline layer of protection for all interactions.



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